MIL-S-36894A
7 July 1978
SUPERSEDING
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MILITARY SPECIFICATION

SYRINGE, DENTAL IMPRESSION MATERIAL
AND REPLACEMENT PARTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a dental syringe with accessory parts, suitable for injecting light-bodied impression material into prepared teeth.

1.2 Classification. The syringe and accessory parts shall be of the following types as specified (see 6.1):

Type I - Syringe, complete (with accessories).
Type II - Tips, curved.
Type III - Accessory kit, with straight tips and "O"-rings.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-C-186 - Containers, Packaging and Packing For Drugs, Chemicals, and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-116 - Preservation-Packaging, Methods Of.

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24278 - Syringe, Dental Impression Material.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC. (USP)

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Materials. The materials and colors of the components of the complete syringe and accessory parts shall be as specified in table I. Materials and properties not specified shall be entirely suitable for the intended purpose.

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TABLE I. Materials and colors.

Component	Material	Color
Syringe body and piston	Rigid Thermoplastic acetal resin	White
"O"-ring	Synthetic rubber	Black
Cap nut	Free cutting brass, nickel plated	-
Tips		
Curved	Semi-rigid cellulose acetate butyrate	Clear or natural
Straight	High density polyethylene	Clear or natural

3.1.1 Cleaning brush. The filaments of the cleaning brush shall be fabricated from nylon. As an alternate, the bristles of the brush shall be fabricated from undyed, natural hog hair. The wire shall be brass or steel made corrosion-resistant by zinc or other suitable coating.

3.1.2 Lubricant. The lubricant for the "O"-rings shall be U.S.P. white petrolatum, in accordance with the tests, standards, and requirements of the United States Pharmacopeia. The immediate container shall be a collapsible metal tube conforming to PPP-C-186. The tube shall contain a minimum of one-eighth ounce.

3.2 Style, design, and dimensions. Style, design, and dimensions of the type I complete syringe with accessory parts, the type II curved replacement tips, and the type III straight replacement tips and replacement "O"-rings shall be in conformance with Defense Personnel Support Center (DPSC) Drawing No. 24278.

3.2.1 Type I syringe, complete (with accessories). Type I complete syringe shall consist of the following: one assembled syringe (body, piston, "O"-ring, and cap nut), one straight and one curved tip, one tube of lubricant, one cleaning brush, one extra "O"-ring, and a set of instructions for use.

3.2.2 Type II tips, curved. Type II curved tips shall be suitable replacements for use with the type I syringe.

3.2.3 Type III accessory kit, with straight tips and "O"-rings. The type III accessory kit with straight tips and "O"-rings shall be suitable replacements for use with the type I syringe.

3.3 Performance. The syringe shall be satisfactory for use in injecting light-bodied, rubber-base impression material, without leakage and without clogging, when tested as specified in 4.3.1.

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3.4 Corrosion resistance. The cap nut shall meet the corrosion resistance test as specified in 4.3.2.

3.5 Finish. Surfaces of the syringe body and piston shall be smoothly finished. Components of syringe shall be free of flash and burrs. The metal cap nut shall have a mirror or highly polished finish, except for the knurled portion. Surfaces of "O"-rings and tips shall be smooth. The syringe body and piston shall provide a smooth operational fit.

3.6 Marking. The type I syringe body shall be legibly and permanently marked with the name or registered trademark of the contractor. Syringes of foreign (other than U.S.) manufacture shall also be marked with the name of the country of origin.

3.7 Instructions. Adequate and accurate instructions for assembly, use, disinfection, and cleaning of the syringe shall be included with each type I complete syringe with accessory parts.

3.8 Workmanship. Workmanship shall be first class throughout. The complete syringe with accessory parts, and replacement parts, shall be clean and free from defects which detract from their appearance or may impair their serviceability.

3.9 Quantity. Each unit of issue shall consist of the components listed in table II.

TABLE II. Components and quantity in unit of issue.

Classification	Component	Quantity
Type I	Complete syringe	1
	Body	1
	Piston	1
	"O"-ring	1
	Cap nut	1
	Accessory parts	
	Curved tip	1
	Straight tip	1
	"O"-ring	1
	Cleaning brush	1
	Tube of lubricant	1
	Instruction phamphlet	1
Type II	Tips, curved	6
Type III	Accessory kit	
	Straight tips	12
	"O"-rings	3

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3.10 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, fit and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the components, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/ material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. One package containing the complete syringe with accessory parts, the package of curved replacement tips, and the accessory kit of straight replacement tips and "O"-rings, as applicable, shall be examined for defects including, but not limited to, those listed in table III.

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TABLE III. Classification of defects.

Categories	Defect
Critical	None defined.
Major A:	
101	Type not as specified.
102	Any mating parts of syringe fail to fit properly.
103	Syringe body not threaded properly to accept cap nut.
104	Style and design of any component of complete syringe assembly not as specified.
Major B:	
151	Any prescribed dimension or dimensional tolerance does not insure correct fit of mating components on syringe.
Minor A:	
201	Finish not as specified.
202	Twisted metal wire of cleaning brush fails to grip nylon filaments securely.
203	Disposable tips not semi-rigid and clear or natural, as applicable.
204	Instructions for proper assembly, use, and cleaning of the syringe either missing, inadequate, inaccurate, or illegible.
205	Identification marking on Type I missing, inadequate, inaccurate, or illegible.
206	Quantities not as specified for applicable type.
Minor B:	
251	Any dimension (except 151) not as specified.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table IV. Unit of product for examination purposes shall be one package containing complete syringe with accessory parts, one package of curved replacement tips, or one accessory kit of straight replacement tips and "O"-rings, as applicable.

TABLE IV. Examination criteria.

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

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4.2.2 Testing. Syringes shall be tested as specified in table V and 4.3. Inspection levels and acceptable quality levels shall be as listed in table V. The unit of product for testing shall be as indicated in the table.

TABLE V. Test criteria.

Characteristic	Require-ment	Test procedure	Unit of product	Inspection level	AQL
Performance	3.3	4.3.1	One syringe	S-1	1.0
Corrosion resistance	3.4	4.3.2	One cap nut	S-2	1.0

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Performance. The syringe shall be properly assembled, and then partially filled with light-bodied, rubber-base impression material conforming to NSN 6520-00-764-2262. The syringe shall then be slowly emptied of the impression material. Observation shall be made for leakage, clogging, and lack of uniform flow of the material, any of which shall render the syringe unsuitable for use.

4.3.2 Corrosion resistance. The cap nut of the syringe shall pass the boil test, as specified in GG-I-526.

4.4 Packaging inspection. The inspection of the packaging shall be in accordance with the group A quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing one syringe with accessory parts, type I, or one package containing six tips, type II, or one accessory kit, type III, as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging. Each unit shall be packaged in accordance with MIL-P-116, method III, as specified below:

5.1.1.1.1.1 Type I. Each unit shall be packaged in a box of appropriate size and design, constructed in accordance with PPP-B-566 or PPP-B-676. The "O"-ring and disposable tip shall be packaged in a plastic box or plastic bag of appropriate size and design. Box holding the complete unit shall be designed with interior fittings. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.1.1.1.2 Type II and Type III. Each unit shall be packaged in a plastic box of appropriate size, with a hinged cover. Closure shall be adequate to prevent accidental opening.

5.1.1.1.2 Intermediate packaging. Twelve units of type I or type II or fifty units of type III shall be packaged in a box of appropriate size, constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Level C. Units shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Two hundred and eighty-eight units of type I or one hundred and forty-four units of type II or eight hundred units of type III shall be packed in an exterior container designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Two hundred and eighty-eight units of type I or one hundred and forty-four units of type II or eight hundred units of type III shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Each unit, intermediate package, and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type specified (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-469-3497	SYRINGE, DENTAL IMPRESSION MATERIAL, Plastic, Thumb Rest Handle
6520-00-469-3498	TIP, DENTAL IMPRESSION MATERIAL SYRINGE, Plastic, Disposable, Curved, 6s
6520-00-890-2077	ACCESSORY KIT, DENTAL IMPRESSION MATERIAL SYRINGE

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6520-1450

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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